

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe(include = 'all')` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	N.
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	N.
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	N.
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	N.
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	N.
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	N.
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	N.

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the
 - `Review Rating` column using the median rating of each product category.
 - `df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))`
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used`
 - were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected Python script to SQL Server and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in SQL Server to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

```
select gender, sum(purchase_amount) as total_revenue
from customer
group by gender
order by sum(purchase_amount)
```

	gender	total_revenue
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

```
select customer_id, purchase_amount
from customer
where discount_applied = 'Yes' and purchase_amount >= (select AVG(purchase_amount) from customer)
```

	customer_id	purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

```
select top 5 item_purchased, round(avg(review_rating),2) as "Average Product Review"
from customer
group by item_purchased
order by avg(review_rating) desc
```

	item_purchased	Average Product Review
1	Gloves	3,86
2	Sandals	3,84
3	Boots	3,82
4	Hat	3,8
5	Skirt	3,78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

```
select shipping_type, round(avg(purchase_amount),2) as "Average Purchase Amount"
from customer
where shipping_type in ('Standard', 'Express')
group by shipping_type
order by avg(purchase_amount)
```

	shipping_type	Average Purchase Amount
1	Standard	58
2	Express	60

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

```
select
  subscription_status,
  count(*) as "Number of customers",
  round(avg(purchase_amount), 2) as "Total spend",
  sum(purchase_amount) as "Total Revenue"

from customer
group by subscription_status
order by avg(purchase_amount) desc
```

	subscription_status	Total spend	Total Revenue	Number of customers
1	Yes	59	62645	1053
2	No	59	170436	2847

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

```
select top 5
  item_purchased,
  count(*) as total_purchases,
  sum(case when discount_applied = 'Yes' then 1 else 0 end) as discounted_purchases,
  cast(100.0 * sum(case when discount_applied = 'Yes' then 1 else 0 end) / count(*) as
decimal (5,2)) as "discount_percentage"
from customer
group by item_purchased
order by discount_percentage desc;
```

	item_purchased	total_purchases	discounted_purchases	discount_percentage
1	Hat	154	77	50.00
2	Sneakers	145	72	49.66
3	Coat	161	79	49.07
4	Sweater	164	79	48.17
5	Pants	171	81	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

```
select
  customer_segment,
  count(*) as "Number of customers"
from (
  select
    case
      when previous_purchases = 1 then 'New'
      when previous_purchases <=10 then 'Returning'
      else 'Loyal'
    end as customer_segment
```

```

    from customer
  ) as segments
group by customer_segment
order by count(*) desc

```

	customer_segment	Number of customers
1	Loyal	3116
2	Returning	701
3	New	83

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

```

with item_counts as (
select
  category,
  item_purchased,
  count(*) as total_orders,
  row_number() over (partition by category order by count(*) desc) as item_rank
from customer
group by category, item_purchased
)

```

```

select item_rank, category, item_purchased, total_orders
from item_counts
where item_rank <= 3
order by category, item_rank

```

	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Belt	161
3	Accessories	Sunglasses	161
4	Clothing	Blouse	171
5	Clothing	Pants	171
6	Clothing	Shirt	169
7	Footwear	Sandals	160
8	Footwear	Shoes	150
9	Footwear	Sneakers	145
10	Outerwear	Jacket	163
11	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

```

select
  case
    when previous_purchases > 5 then 'Repeat Buyer'
    else 'Regular Buyer'
  end as buyer_type,
  count(*) as total_customers,
  sum(case when subscription_status = 'Yes' then 1 else 0 end) as subscribers,

```

```

cast(100.0 * sum(case when subscription_status = 'Yes' then 1 else 0 end) / count(*) as
decimal(5,2)) as subscriber_percentage
from customer
group by
case
when previous_purchases > 5 then 'Repeat Buyer'
else 'Regular Buyer'
end;

```

	buyer_type	total_customers	subscribers	subscriber_percentage
1	Regular Buyer	424	95	22.41
2	Repeat Buyer	3476	958	27.56

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

```

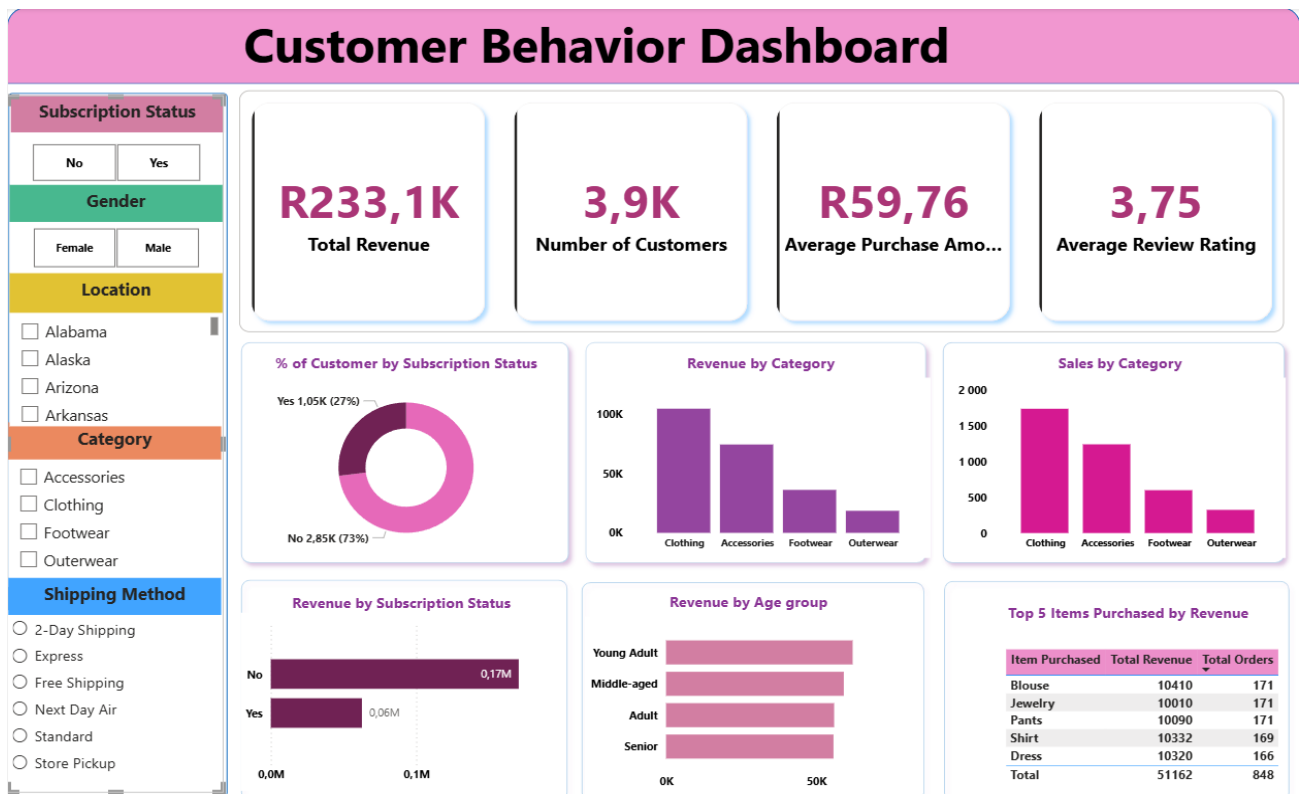
select age_group,
sum(purchase_amount) as total_revenue
from customer
group by age_group
order by total_revenue desc

```

	age_group	total_revenue
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.